

# Determining the Crystalline Structure of Metals through X-Ray Diffraction

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## Abstract

This experiment set out to measure X-ray diffraction patterns of various solid samples in order to describe their crystalline structure level. A pre-calibrated Siemen D5000 X-Ray diffractometer was used to characterize the structure of Copper, Nickel (Ni), CuNi alloy, Titanium (Sn), Lead (Pb), PbSn alloy and two mystery samples A and B. Bragg's law was used to obtain the distances of various diffraction planes, which was related to the Miller indices of the planes. It was found that Copper, Nickel and the alloys are Cubic system with FCC Bravais lattices ((and a lattice parameter of [...])); Tin, Lead and the alloys are Tetragonal systems with [P/I] lattices ((and parameters of )); sample A is a [...] system with a [...] lattice whereas B is a [...] system with [...] lattice.

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# 1 Introduction

A solid can serve as a diffraction grating for X-rays of wavelengths similar to the lengths of inter-atomic distances. Similarly to how light diffracts, X-rays can also diffract off of atoms. Atoms within solids will arrange themselves in symmetries, formally called cells, given by one of one of fourteen Bravais lattices, with each lattice grouped into one of seven crystal systems. These Bravais lattices govern along which diffraction planes atoms may align and Miller indices are conventionally used to index the various planes. Appendix 6.1 shows the different Bravais lattices. If an X-ray is sent to a solid with an angle  $\theta$  such that constructive interference occurs at a detector placed  $2\theta$  with respect to the incident ray, a peak in intensity of photons may be observed. From this, one can obtain a relation between the inter-planar distance  $d$ , the incident angle  $\theta$ , and the wavelength  $\lambda$ .

$$n\lambda = 2d \sin(\theta) \tag{1}$$

with  $n = 1$  for our purposes. Equation 1 is formally called Bragg's Law.

Using these observed peaks and (1), the distance  $d$  between the parallel diffraction planes can be obtained and related to the Bravais lattice and Miller indices by various equations. For a cubic system, this relation takes the form of

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2} \tag{2}$$

where  $h, k, l$  are the Miller Indices and  $a$  is the length of the cube edge. Similar equations for different crystal systems are given in 6.1.

## 2 Methods

### 2.1 Instrument and Data Acquisition

The apparatus consisted of a pre-calibrated Siemens D5000 diffractometer. The X-ray tube used Copper  $K_\alpha$  radiation. Two sets of filters of 0.6 mm were present on the X-ray tube

and the detector. During data acquisition, the diffractometer was set to 40 kV and 40 mA. A computer connected to the apparatus was used to specify the measurement parameters and obtain the data from the instrument. A picture of the instrument is shown in Figure 1

Data acquisition consisted in mounting the sample on the mounting plate, opening the parameter file in the computer software and executing the job. Once the job is executed, the detector rotates about the mounting plate to be at an angle of  $2\theta_0 = 5.00^\circ$  with respect to the incident rays and the mounting plate rotates about itself to make an angle of  $\theta_0$ . The detector and mounting plate rotate with angle increments of  $2\Delta\theta = 0.05$  and  $\Delta\theta$ , respectively, with 2.00 s at each step until reaching  $2\theta = 110.00^\circ$ .

Measurements were taken for samples of Copper, Nickel, Tin, Lead; 25%, 50% and 75% alloys of Copper-Nickel and Tin-lead, and two mystery samples A and B. Three trials were initially done for Copper and Nickel, and two trials were done for the other samples. It was found that the uncertainties associated to statistical fluctuations of two measurements were smaller than that of the standard deviation of gaussians fitted about the intensity peaks.

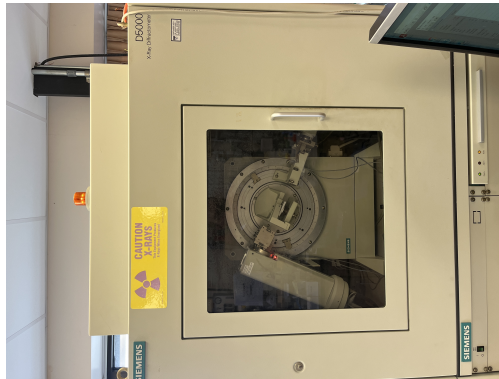


Figure 1: Picture of the Siemens D5000 diffractometer used for the experiment.

## 2.2 Data Analysis

For a given sample, the acquired data is used to plot intensity against  $2\theta$ . From the plots, local maxima of intensity are obtained and gaussians are fitted against individual peaks. Noise is accounted for by taking an average of the intensities where there are no peaks, and subtracting it from the peaks. The Gaussians are then summed and plotted. From the means of the gaussians the angle of diffraction is obtained and used to find the inter-planar distance

(for that given peak), using equation 1. Finally, a mean inter-planar distance  $d_{mean}$  is taken between all the runs for a given sample. Uncertainties were first obtained in  $2\theta$ , from the standard deviation of the associated gaussian. This was then propagated to  $d$  and to  $d_{mean}$ . Consider the set of equation in Appendix 6.1 relating  $d$  and the Miller indices. For every  $d$  spacing, we want to find what the sum of the Miller indices are. These will always be integers, so the goal is to make a guess on  $h, k, l$ , find the lattice constant(s), and check for consistency. Notice that  $d$  is maximised when  $h, k, l$  are a minimum, and that not all integer for  $h, k, l$  are allowed. By guessing certain values of  $h, k, l$ , we can isolate the lattice parameters, take the ratios of the spacings with the latter, and if consistent these ratios should be integers. The ratios may also be multiplied by a small integer to become integer if they aren't already. For tetragonal and hexagonal system, this process is repeated twice for each lattice parameter. For the Mystery samples and alloys, an extra guess is made on the crystal system before attempting to decipher the Miller indices. After the Miller indices are obtained, the Bravais lattice can be found using selection rules: certain Bravais lattices will allow only a specific set of diffraction planes.

### 3 Results

Figure 2 shows the summed gaussians fitted to the diffraction pattern for the Cu sample. Notice the peak of the gaussians is lower than that of the data, this is to account for noise. From the means of the gaussians and Bragg's law the inter-planar distances were found, as shown in Tables ??

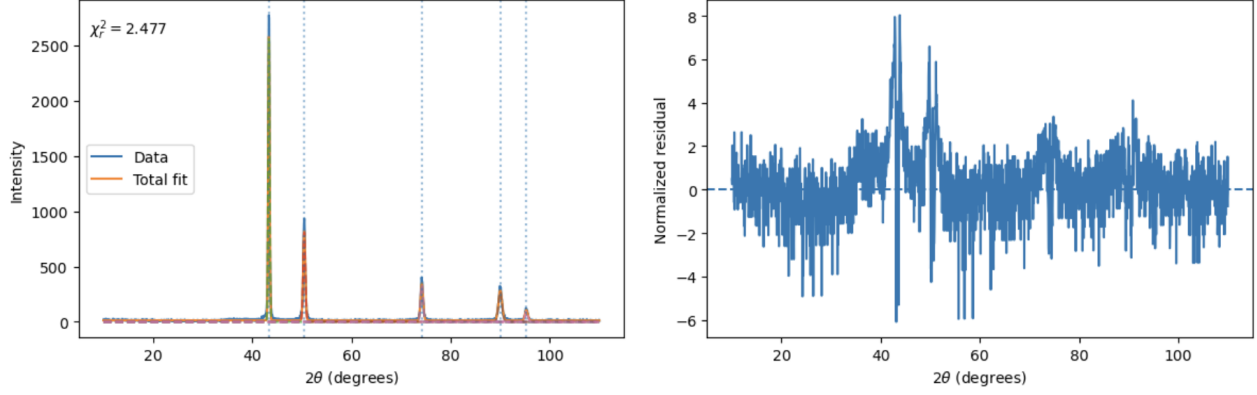


Figure 2: Plot of intensity against  $2\theta$  for the Copper sample. the summed gaussians (orange) are fitted to the data (blue).

| $2\theta$ (deg)               | $d_{mean}$ (Å)                |
|-------------------------------|-------------------------------|
| $43.3664 \pm 8 \cdot 10^{-4}$ | $2.08485 \pm 4 \cdot 10^{-5}$ |
| $50.489 \pm 2 \cdot 10^{-3}$  | $1.80619 \pm 6 \cdot 10^{-5}$ |
| $74.205 \pm 3 \cdot 10^{-3}$  | $1.27694 \pm 5 \cdot 10^{-5}$ |
| $90.017 \pm 4 \cdot 10^{-3}$  | $1.08921 \pm 4 \cdot 10^{-5}$ |
| $95.263 \pm 7 \cdot 10^{-3}$  | $1.04260 \pm 6 \cdot 10^{-5}$ |

Table 1: Experimental and theoretical lattice parameters for Cu, Ni pure samples and alloys. The theoretical values for the alloys are calculated using Vegard's law.

| $1/d^2$ (Å <sup>-2</sup> )     | $\frac{1/d^2}{a^2} \cdot 3$   | $h^2 + k^2 + l^2$ | index |
|--------------------------------|-------------------------------|-------------------|-------|
| $0.230064 \pm 8 \cdot 10^{-6}$ | $3.00 \pm 0.00$               | 3                 | (111) |
| $0.30653 \pm 2 \cdot 10^{-5}$  | $3.9971 \pm 3 \cdot 10^{-4}$  | 4                 | (200) |
| $0.61329 \pm 4 \cdot 10^{-5}$  | $7.9971 \pm 6 \cdot 10^{-4}$  | 8                 | (220) |
| $0.84291 \pm 6 \cdot 10^{-5}$  | $10.9913 \pm 8 \cdot 10^{-4}$ | 11                | (311) |
| $0.9200 \pm 1 \cdot 10^{-4}$   | $11.996 \pm 1 \cdot 10^{-3}$  | 12                | (222) |

Table 2: Experimental and theoretical lattice parameters for Cu, Ni pure samples and alloys. The theoretical values for the alloys are calculated using Vegard's law.

The fitted gaussian sums with residuals for the Cu and Ni samples and alloys are shown.

In figure 5, the residuals for the Ni samples are shown, and from their numerical results in table ?? we can observe that random error from trial to trial is minimal compared with noise error provided by the Gaussian fits. The fitted gaussian sums with residuals for the Cu and Ni samples and alloys are shown in the same appendix, in figure 6. From these, the inter-planar distances were calculated and are shown in table 3. These were plotted against expected results for all 3 types of cubic Bravais lattices, and the best fit was found to be the Face-Centered Cubic (FCC) lattice, as shown below in figure 3. The lattice parameters were then calculated and are shown in table 3.

| Sample                            | $a_{\text{exp}}$ (Å) | $a_{\text{theory}}$ (Å) | $\Delta a$ (Å)       |
|-----------------------------------|----------------------|-------------------------|----------------------|
| Ni                                | $3.5223 \pm 0.0001$  | 3.5238                  | $-0.0015 \pm 0.0001$ |
| Cu                                | $3.6126 \pm 0.0001$  | 3.6149                  | $-0.0023 \pm 0.0001$ |
| Cu <sub>25</sub> Ni <sub>75</sub> | $3.5415 \pm 0.0002$  | 3.5467                  | $-0.0052 \pm 0.0002$ |
| Cu <sub>50</sub> Ni <sub>50</sub> | $3.5629 \pm 0.0003$  | 3.5696                  | $-0.0067 \pm 0.0003$ |
| Cu <sub>75</sub> Ni <sub>25</sub> | $3.5864 \pm 0.0002$  | 3.5924                  | $-0.0060 \pm 0.0002$ |

Table 3: Experimental and theoretical lattice parameters for Cu, Ni pure samples and alloys. The theoretical values for the alloys are calculated using Vegard’s law .

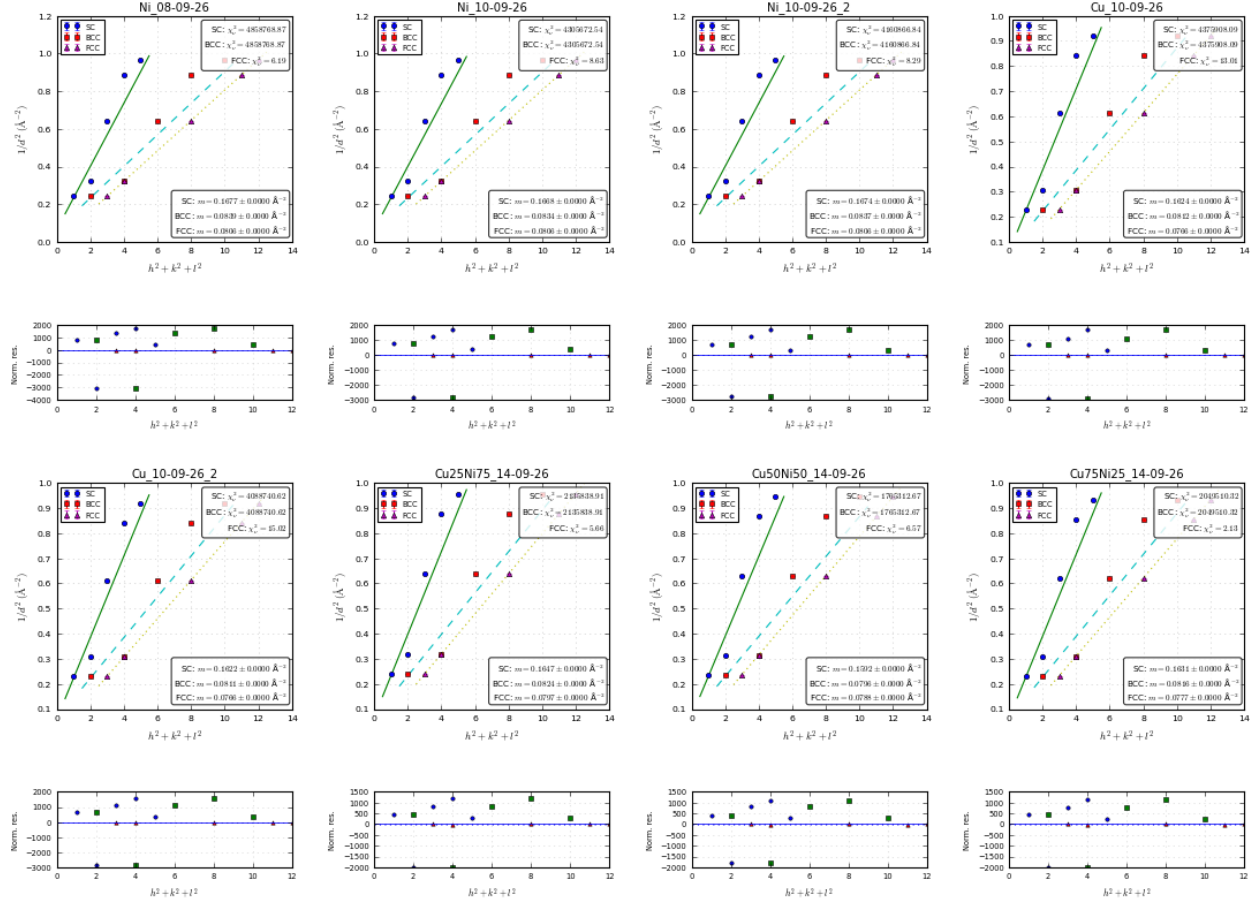


Figure 3: Determination of Cubic crystal structure and lattice parameter for Cu and Ni.

## 4 Discussion

The results for the Cu and Ni pure samples and alloys are consistent with the literature values, with the experimental values being within 0.2% of the theoretical values. The small discrepancies can be attributed to background noise in the acquired data. The determination of the Face-Centered Cubic (FCC) Bravais lattice for these samples is also consistent with known crystal structures for Cu and Ni. While running multiple trials does not reduce error much, it could still be worth it, as random error is of similar order as the noise error. The results for the Pb, Sn, and PbSn alloys are still to be determined, as well as the mystery samples A and B.

## 5 Conclusion

## 6 Appendix

### 6.1 Inter-planar distance, Bravais Lattices and Miller indices

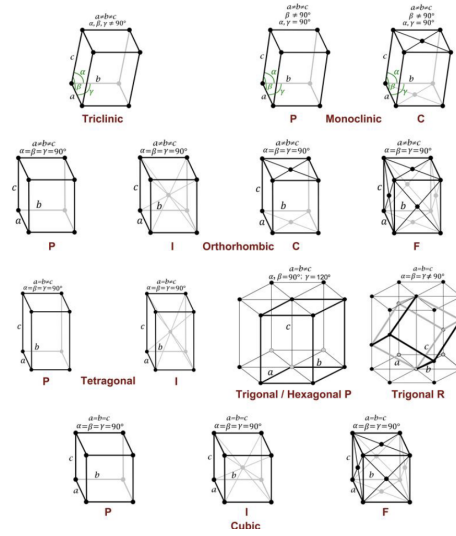


Figure 4: Fourteen Bravais lattices and their associated seven groups.

For the following equations:  $d$  is the inter planar distance,  $a, b, c$  the inter-atomic distances,  $h, k, l$  the indices for a given Miller index  $(hkl)$ .

Cubic:

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

Tetragonal:

$$\frac{1}{d^2} = \frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}$$

Hexagonal:

$$\frac{1}{d^2} = \frac{4}{3} \left( \frac{h^2 + k^2 + hk}{a^2} \right) + \frac{l^2}{c^2}$$

Orthorhombic:

$$\frac{1}{d^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

Vegard's law:

$$a_{\text{alloy}} = xa_{Cu} + (1 - x)a_{Ni}$$

## 6.2 Supplementary Figures and Tables

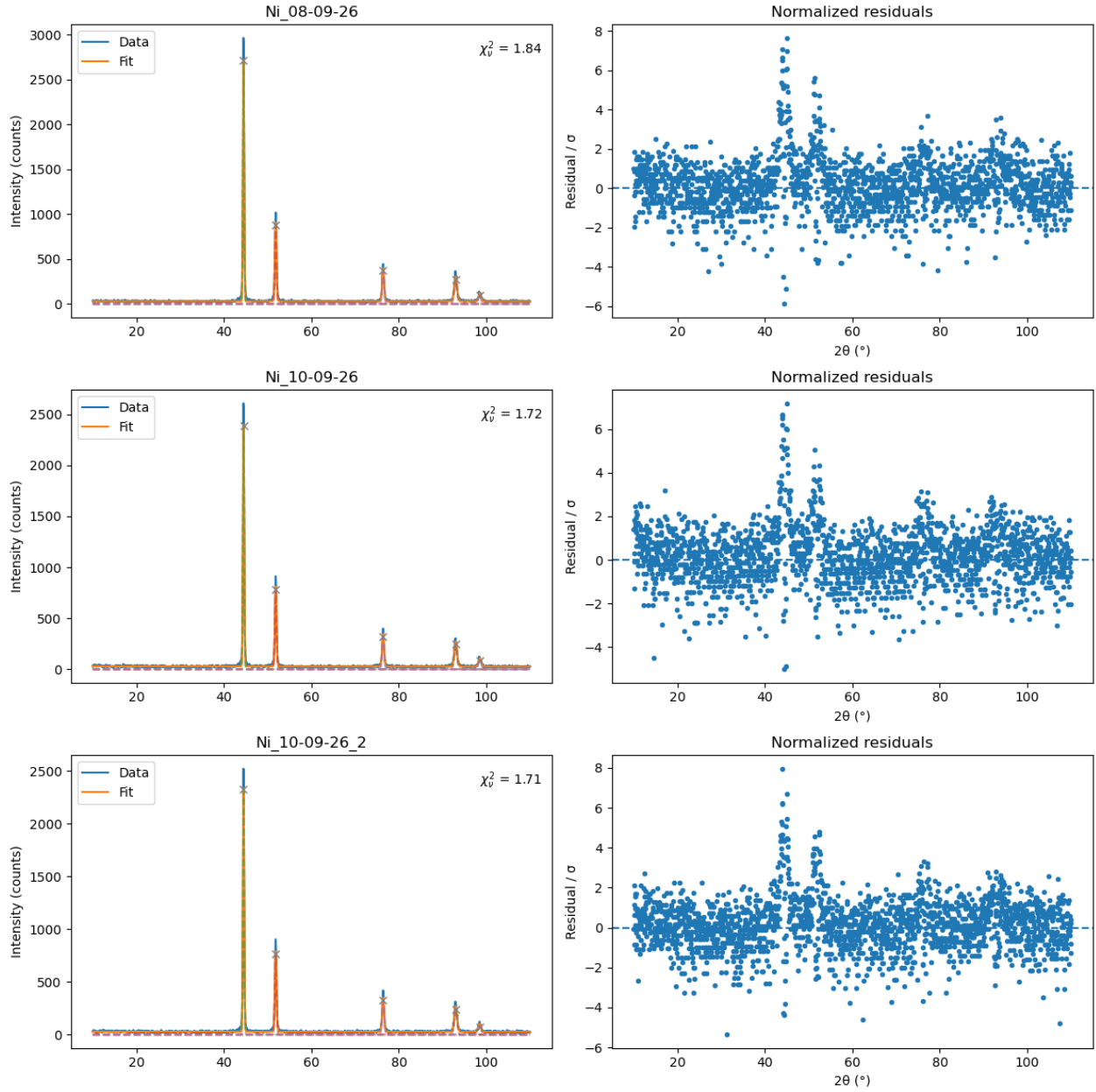


Figure 5: Gaussian sum fits of all pure Ni trials with residuals

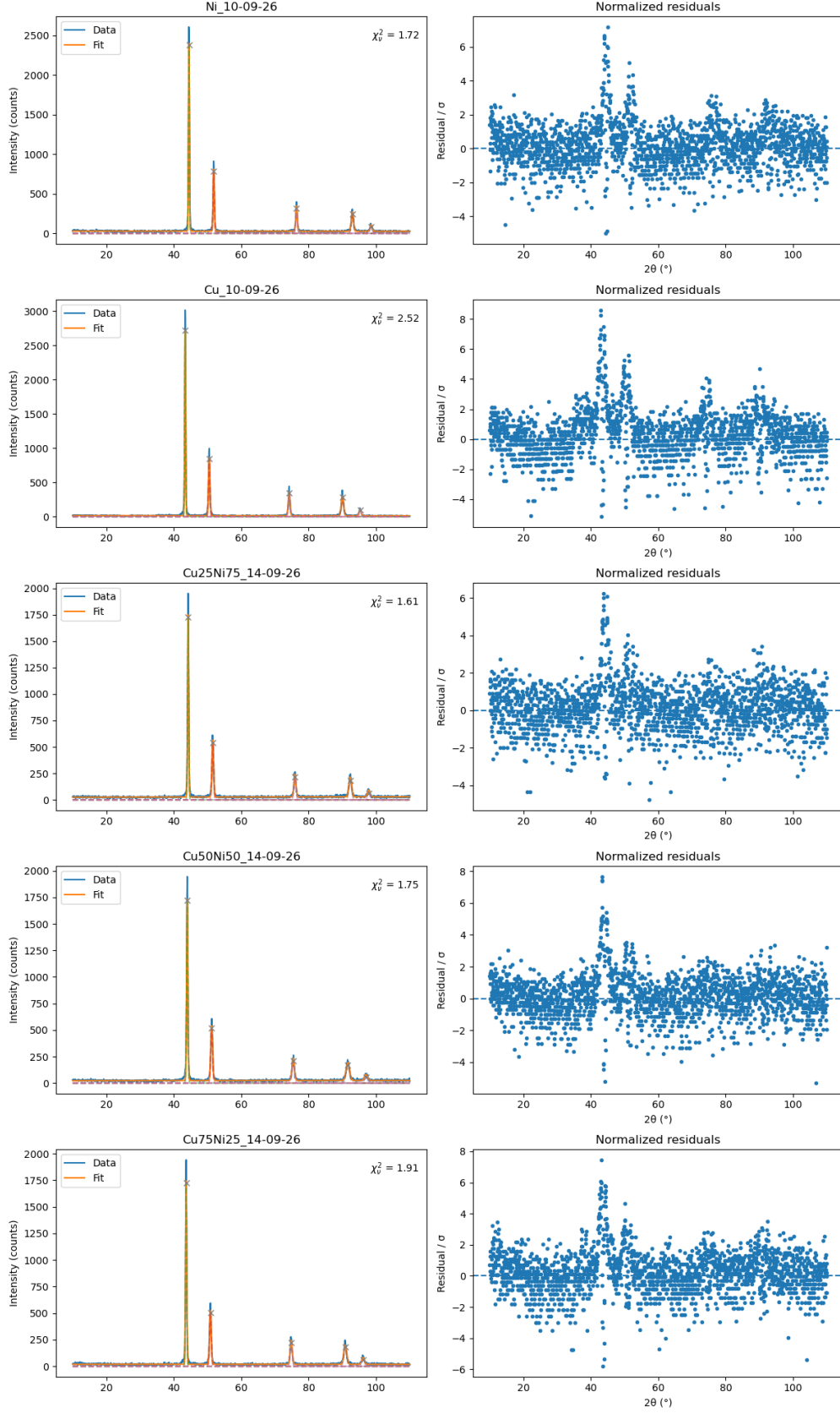


Figure 6: Gaussian sum fits with residuals of the Cu and Ni pure and alloy samples

| Ni trial 1                        |                      |                       | Ni trial 2                        |                      |                       |
|-----------------------------------|----------------------|-----------------------|-----------------------------------|----------------------|-----------------------|
| Peak                              | $2\theta$ (°)        | $d$ (Å)               | Peak                              | $2\theta$ (°)        | $d$ (Å)               |
| 1                                 | $44.4819 \pm 0.0011$ | $2.03512 \pm 0.00005$ | 1                                 | $44.4843 \pm 0.0011$ | $2.03502 \pm 0.00005$ |
| 2                                 | $51.8330 \pm 0.0023$ | $1.76246 \pm 0.00007$ | 2                                 | $51.8331 \pm 0.0024$ | $1.76245 \pm 0.00008$ |
| 3                                 | $76.3993 \pm 0.0041$ | $1.24563 \pm 0.00006$ | 3                                 | $76.3933 \pm 0.0045$ | $1.24571 \pm 0.00006$ |
| 4                                 | $92.9684 \pm 0.0059$ | $1.06221 \pm 0.00005$ | 4                                 | $92.9701 \pm 0.0062$ | $1.06220 \pm 0.00005$ |
| 5                                 | $98.4832 \pm 0.0109$ | $1.01694 \pm 0.00008$ | 5                                 | $98.5025 \pm 0.0127$ | $1.01679 \pm 0.00010$ |
| Ni trial 3                        |                      |                       | Cu trial 1                        |                      |                       |
| Peak                              | $2\theta$ (°)        | $d$ (Å)               | Peak                              | $2\theta$ (°)        | $d$ (Å)               |
| 1                                 | $44.4810 \pm 0.0012$ | $2.03517 \pm 0.00005$ | 1                                 | $43.3661 \pm 0.0011$ | $2.08487 \pm 0.00005$ |
| 2                                 | $51.8307 \pm 0.0024$ | $1.76253 \pm 0.00008$ | 2                                 | $50.4888 \pm 0.0024$ | $1.80618 \pm 0.00008$ |
| 3                                 | $76.4023 \pm 0.0045$ | $1.24558 \pm 0.00006$ | 3                                 | $74.2095 \pm 0.0043$ | $1.27687 \pm 0.00006$ |
| 4                                 | $92.9767 \pm 0.0062$ | $1.06214 \pm 0.00005$ | 4                                 | $90.0228 \pm 0.0054$ | $1.08915 \pm 0.00005$ |
| 5                                 | $98.4665 \pm 0.0135$ | $1.01707 \pm 0.00010$ | 5                                 | $95.2553 \pm 0.0097$ | $1.04266 \pm 0.00008$ |
| Cu trial 2                        |                      |                       | Cu <sub>25</sub> Ni <sub>75</sub> |                      |                       |
| Peak                              | $2\theta$ (°)        | $d$ (Å)               | Peak                              | $2\theta$ (°)        | $d$ (Å)               |
| 1                                 | $43.3667 \pm 0.0011$ | $2.08484 \pm 0.00005$ | 1                                 | $44.2433 \pm 0.0015$ | $2.04554 \pm 0.00006$ |
| 2                                 | $50.4882 \pm 0.0024$ | $1.80620 \pm 0.00008$ | 2                                 | $51.5406 \pm 0.0034$ | $1.77176 \pm 0.00011$ |
| 3                                 | $74.2000 \pm 0.0043$ | $1.27701 \pm 0.00006$ | 3                                 | $75.9226 \pm 0.0065$ | $1.25226 \pm 0.00009$ |
| 4                                 | $90.0104 \pm 0.0057$ | $1.08927 \pm 0.00005$ | 4                                 | $92.3239 \pm 0.0085$ | $1.06793 \pm 0.00008$ |
| 5                                 | $95.2715 \pm 0.0098$ | $1.04253 \pm 0.00008$ | 5                                 | $97.7855 \pm 0.0166$ | $1.02232 \pm 0.00013$ |
| Cu <sub>50</sub> Ni <sub>50</sub> |                      |                       | Cu <sub>75</sub> Ni <sub>25</sub> |                      |                       |
| Peak                              | $2\theta$ (°)        | $d$ (Å)               | Peak                              | $2\theta$ (°)        | $d$ (Å)               |
| 1                                 | $43.9763 \pm 0.0016$ | $2.05734 \pm 0.00007$ | 1                                 | $43.6499 \pm 0.0015$ | $2.07197 \pm 0.00007$ |
| 2                                 | $51.2219 \pm 0.0036$ | $1.78204 \pm 0.00012$ | 2                                 | $50.8455 \pm 0.0035$ | $1.79434 \pm 0.00011$ |
| 3                                 | $75.4112 \pm 0.0071$ | $1.25948 \pm 0.00010$ | 3                                 | $74.8037 \pm 0.0062$ | $1.26819 \pm 0.00009$ |
| 4                                 | $91.6044 \pm 0.0098$ | $1.07443 \pm 0.00009$ | 4                                 | $90.8289 \pm 0.0083$ | $1.08157 \pm 0.00008$ |
| 5                                 | $97.0016 \pm 0.0200$ | $1.02849 \pm 0.00016$ | 5                                 | $96.1327 \pm 0.0160$ | $1.03546 \pm 0.00013$ |

Table 4: Measured X-ray diffraction peak angular positions and corresponding interplanar spacings ( $d$ ) for the Ni, Cu, and Cu–Ni alloy samples.